

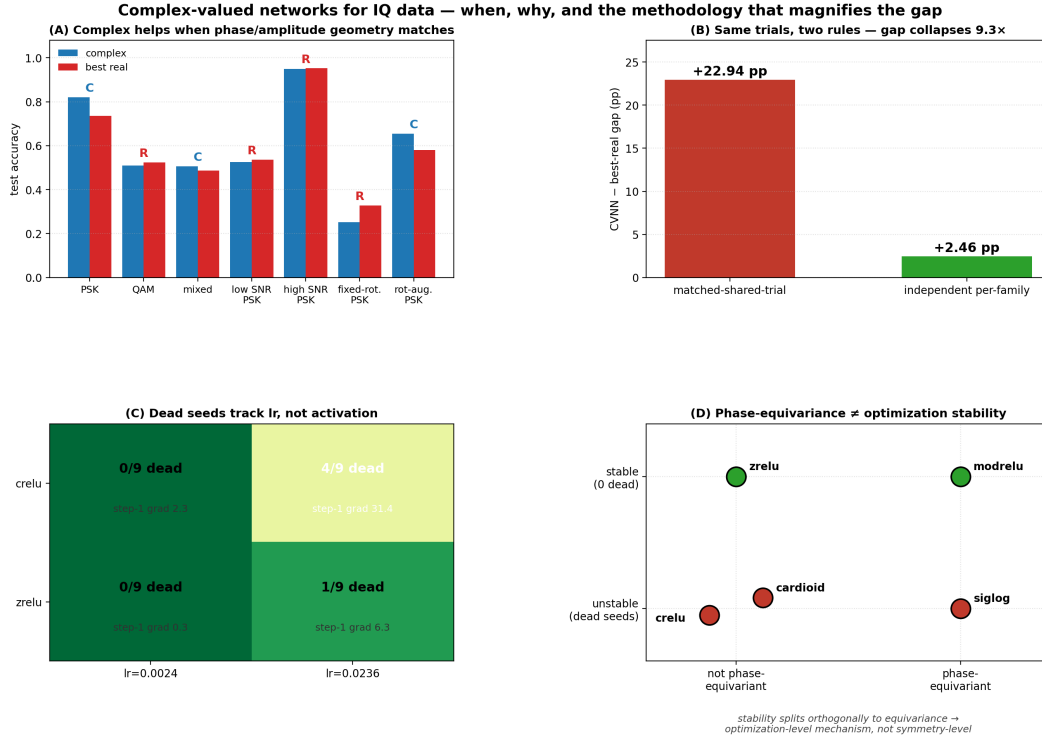


Figure 1: The paper at a glance. (A) Complex-valued networks win on phase-bearing modulations (PSK, mixed) but lose to real magnitude on QAM and to matched-real at low SNR or under fixed carrier-phase rotation; the advantage is conditional on representation/SNR/invariance assumptions. (B) The 22.94 pp `crelu` gap on a 3-class RadioML subset collapses to 2.46 pp under independent per-family hyperparameter selection — a 9.3× change driven entirely by the selection rule, not the data. (C) A 2×2 lr-vs-activation disambiguation (Section 3.6.7) shows the dead-seed threshold follows the learning rate, not the activation. (D) Phase equivariance of the activation does not predict optimization stability: the four quadrants are populated, so the robustness asymmetry is an optimization-level effect, not a symmetry-level one.

Complex spaces as representation, geometry, and optimization: stress-testing architectural priors and training robustness in complex-valued neural networks

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Abstract

Complex-valued neural networks (CVNNs) are usually motivated by phase-bearing data, but the phrase “complex helps” hides several different mechanisms: the coordinate representation of the input, the algebraic prior imposed by complex multiplication, and the optimization dynamics induced by coupled real/imaginary parameters. We study these mechanisms at a scale where they can be separated rather than conflated. Synthetic representation stress tests show that the complex advantage is conditional: PSK-only IQ data favors the complex model (0.821 vs 0.333 for magnitude-only), QAM-only inverts that ranking, mixed PSK+QAM gives a small complex edge, and an unseen global phase rotation collapses every coordinate-dependent model unless rotation augmentation is added. Complex-valued models are therefore best framed as a useful inductive bias when the task’s phase/amplitude geometry matches the model. A feasibility-scale quantum wavefunction pilot extends the same question outside RF: phase-aware models solve momentum-from-wavefunction classification perfectly while magnitude-only remains at chance (1.000 vs 0.250), but a 1-D potential-inverse task is best matched by an explicit real Cartesian view (0.462 vs 0.431 for complex), again emphasizing representation rather than universal complex superiority. The headline empirical finding is methodological: on a 3-class RadioML 2018.01A subset, matched-shared-trial hyperparameter selection produces a 22.94 pp CVNN-vs-real gap that collapses to 2.46 pp under independent per-family selection. Per-step gradient telemetry attributes the difference to a step-1 `head.weight` gradient explosion — 50–80× larger for real baselines than complex — that pushes a fraction of real-baseline seeds into a dead-ReLU region they cannot escape. A 2×2 lr-vs-activation disambiguation localizes the dead-seed threshold primarily to learning rate: at `lr = 0.0024` both `crelu` and `zrelu` produce 0/9 dead seeds, and at `lr = 0.0236` both produce dead seeds (4/9 on `crelu`, 1/9 on `zrelu`); activation modulates sensitivity at high lr but does not by itself trigger the failure mode. We therefore read matched-shared-trial as a robustness stress test, not as the magnitude of the complex advantage. The resulting paper is a scoped mechanism study: it asks when, why, and under what invariance assumptions complex-valued RF models are worth using, and it releases code, manifests, stress-test plots, telemetry traces, and figure scripts for every claim.

1 Concept and feasibility scope

1.1 What this paper claims and does not claim

We make six concrete claims, each at a defensible scale and with explicit reproducibility:

1. **Architectural claim (formal).** `ComplexConv1d` is phase-equivariant in the bias-free case (Section 1); a magnitude-logit head is invariant to global phase rotations of its input features. A real `Conv1d` on stacked (I, Q) channels does not get this for free — it must learn rotation invariance from data via a 2×2 channel mix the complex network gets by construction.
2. **Representation claim.** The complex-vs-real comparison is not meaningful unless we also control the coordinate system presented to real networks. On PSK-only synthetic RF, phase-aware encodings learn while magnitude-only remains at chance; on QAM-only data magnitude becomes competitive; on mixed PSK+QAM the complex edge is small. We therefore report complex, Cartesian real, polar, phase-only, and magnitude-only baselines together (Section 3.4) rather than treating (Re, Im) stacking as the only real baseline.
3. **Application pilot claim.** A 1-D quantum wavefunction pilot (Section 3.5) reproduces the same boundary logic in a non-RF domain. When the label is wavepacket momentum, the information lives in the phase gradient and magnitude-only remains at chance. When the task is inverse potential classification after Schrödinger evolution, the task is learnable above chance but the best model is an explicit real Cartesian view, not the complex model. The pilot supports the application scope of representation stress testing; it does not convert the paper into a broad physics benchmark.
4. **Empirical claim.** Complex-valued networks beat capacity-matched real-valued ones on synthetic IQ modulation classification (Section 3.3) and on a 3-class subset of RadioML 2018.01A [6] (Section 3.6) under matched-budget evaluation, but the magnitude and interpretation of those wins are conditional on representation, SNR, activation, and invariance assumptions. On a task with no temporal phase structure (1-D phase classification, Section 3.2), the complex inductive bias offers no measurable advantage. Both positive and null results are reported.
5. **Optimization claim.** The apparent magnitude of the RadioML win is activation-dependent (Section 3.6.1) because matched-shared-trial selection picks high-lr regimes under some activations that real baselines cannot tolerate. Per-step gradient telemetry (Section 3.6.5) localizes the failure to a step-1 head-gradient explosion that puts a fraction of real-baseline seeds into a dead-ReLU region they cannot escape; the threshold behavior replicates on synthetic data. A 2×2 lr-vs-activation disambiguation (Section 3.6.7) confirms the threshold is primarily lr-driven: at low lr both `crelu` and `zrelu` produce zero dead seeds; at high lr both produce them, with `crelu` more sensitive than `zrelu` (4/9 vs 1/9). Activation choice modulates the magnitude of the step-1 head explosion but does not by itself trigger the failure mode. Phase equivariance of the *layer* is necessary for the architectural prior, but activation-level phase equivariance does not predict the empirical robustness asymmetry — locating the asymmetry one level lower, in (Re, Im) parameter coupling at the optimization level, is what makes the picture self-consistent (Section 4.1).
6. **Methodological claim.** The matched-shared-trial gap (+22.94 pp on `crelu`) and the independent-per-family gap (+2.46 pp on the same data) differ by an order of magnitude. We read matched-shared-trial as a robustness stress test and independent selection as the better point estimate of best achievable performance under separate tuning. Reporting both is part of the contribution; reporting only one risks overstating or erasing a real but small effect.

We do *not* claim:

- That complex arithmetic is universally better than real-valued learning. The RF stress tests deliberately include cases where real magnitude or matched real baselines win. The contribution is to map the boundary of the advantage, not to erase the boundary.
- That CVNNs beat real-valued networks at full RadioML 2018.01A scale (24 modulations $\times$ 26 SNR levels $\times$ 1024-sample length). The infrastructure for that experiment is in place

(`experiments/rf/sweep_radioml.py -preset full`), but the run is deferred. The mechanism analysis is the contribution; scale-up is signposted future work.

- That the inductive bias survives in 2-D regimes (audio STFT, MRI). `ComplexConv2d` is not implemented; the project is 1-D only.
- That the quantum pilot establishes broad physics superiority. It is a deliberately small application probe over 1-D wavefunctions and five potential classes, included to test whether the representation thesis transfers outside RF.
- That CReLU activations generalize across signal-processing settings. The activation ablation (Section 3.6.1) is part of the contribution precisely because activation choice changes the appearance (though not the direction) of CVNN-vs-real gaps.

1.2 Why feasibility-scale is the right scope

A mechanism-level study requires the ability to (a) instrument every parameter’s gradient at every step, (b) run hundreds of training configurations within a working budget, and (c) confirm cross-task generalization with a second data distribution. All three are tractable at the scope chosen — 60 instrumented runs per data source, a re-runnable budgeted sweep, a synthetic stand-in matching the real-data benchmark on architecture and hyperparameter regime. They are not tractable for paper deadlines at $24\text{-class} \times 1024\text{-sample} \times 100\text{-seed}$ scale. We treat the smaller scope as a positive feature: it lets us report what is happening, not just that something is happening.

1.3 Reproducibility as a first-class concern

Every empirical claim above is backed by a committed result manifest that records the Python / PyTorch / OS / device / dtype, every seed used, the exact `git_commit` of the producing code, and a `git_dirty` boolean. CI surfaces newly added dirty manifests so they are reviewed deliberately rather than slipping through. A structural CI guard checks that the activation characterization artifact regenerates without errors and contains all six expected activation rows. The companion repository currently has 139 passing unit and integration tests, with two CUDA-specific tests skipped on the local non-CUDA development machine.

2 Framework

The infrastructure exists to answer the “but what about...?” questions that ad-hoc CVNN comparisons in the literature routinely sidestep.

CVNN library. We implement `ComplexLinear` (matmul + bias-add, deliberately bypassing `F.linear` to sidestep its documented MPS gap), `ComplexConv1d` (thin wrapper over `F.conv1d`, which natively supports complex tensors on CPU/MPS/CUDA), and `ComplexDropout` using a single shared real Bernoulli mask following Trabelsi et al. [7]. Six activations span the Liouville trade-off: CReLU, ZReLU, ModReLU [1], `ComplexCardioid`, `Siglog`, and `ComplexTanh` as the cautionary baseline. Initialization (Glorot [2] and Kaiming [3]) is complex-aware in both rectangular and polar (Rayleigh-magnitude / uniform-phase) forms.

Activation characterization toolkit. For each candidate activation, we render a Cauchy–Riemann residual map over the complex plane, compute summary statistics (median / p95 / max), and measure two gradient norms: an MLP-scaffolded one and an *intrinsic* one ($|\partial L / \partial \bar{z}|$ for $L = \sum |f(z)|^2$ on uniformly sampled z). The intrinsic measure decouples the activation’s contribution from any surrounding linear layers — without it, `tanh`’s 855 MLP-gradient looks $\sim 8\times$ larger than its 108 intrinsic contribution.

Baseline matching. Every claim of “complex beats real” is backed by four families running on the same task with the same shared search space: *naive real-stacked* (real net taking (Re, Im) as 2 channels; $\sim 2\times$ the parameters of the complex layer); *matched-parameter real* (real net sized down so its `numel` equals the complex’s, counting each complex slot as 2); *matched-FLOP real* (sized to match the complex MAdd count); *exact reparameterization* (the block-real $\begin{bmatrix} A & -B \\ B & A \end{bmatrix}$ construction proven to reproduce the complex forward bit-for-bit; a witness, not a baseline to beat).

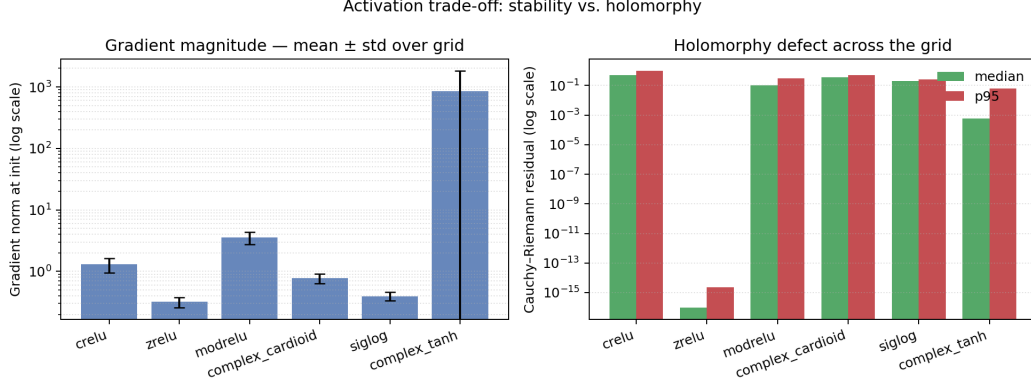


Figure 2: Liouville trade-off, made empirical. *Left*: gradient norm at init for each candidate activation, log scale. *Right*: median and p95 Cauchy–Riemann residual on a $[-3, 3]^2$ grid in $\mathbb{C}$, log scale. `complex_tanh` is locally holomorphic but unbounded (poles at $\pm i\pi/2$ inside the grid); `siglog` is bounded but never holomorphic; `crelu`, `zrelu`, `modrelu`, and `complex_cardioid` sit between, each making a different compromise.

Representation stress tests. For RF synthetic data we add a second axis of control: what information the real network receives. In addition to Cartesian stacking (Re , Im), the stress suite tests polar ($|z|$, $\cos \theta$, $\sin \theta$), phase-only ($\cos \theta$, $\sin \theta$), and magnitude-only $|z|$ views. The same runner probes PSK-only, QAM-only, mixed PSK+QAM, low- and high-SNR regimes, unit-magnitude and unit-power normalization, fixed carrier phase rotation, random-rotation augmentation, and complex activation choice. This turns the research question from “does complex beat real?” into the more falsifiable question “which representation and invariance assumptions explain the gap?”

Budgeted random search. Every model family draws from the *same* shared search space and consumes the same trial $\times$ seed budget. The sweep harness samples the hyperparameter list once and runs it against each family in turn. The primary matched comparison chooses the complex reference trial by mean validation accuracy, then reports every real baseline at that same trial index. Independent family winners are also written as diagnostics but are not the matched-capacity paper table.

3 Findings

3.1 Activation characterization

Section 2 makes the Liouville bind empirically visible. `complex_tanh` has near-zero CR residual at the median — it really is locally holomorphic — but the same metric’s max is 905, because the grid (extent = 3) brackets the first pair of poles at $z = \pm i\pi/2 \approx \pm 1.5708i$. The intrinsic Jacobian mean is ~ 108 . By contrast, `siglog` is bounded ($|f| < 1$ everywhere) and never blows up, but its CR residual is everywhere positive (p95 = 0.25) — it is not holomorphic anywhere, by construction. The remaining four activations sit between, each making a different compromise.

This is not a finding about which activation is best in training — it is a visible measurement of the formal trade-off the project is built around. The choice of activation in the downstream benchmarks (default `crelu`) is consistent across runs and isolated from the architectural variables.

3.2 Synthetic phase classification — null result

After a 16×3 budgeted sweep, the matched shared-trial comparison puts all four families at ~ 76.2 – 76.7% test accuracy with overlapping CIs (Section 3). On 1-D phase classification, **the complex inductive bias does not help**. A 2-D real network on (Re , Im) already has access to all the information the task contains; there is no temporal or geometric structure for $\mathbb{C}$ -multiplication to exploit. The Section 1 of Section 4.1 predicts exactly this — a single-sample phase task does not invoke any sequence-level equivariance.

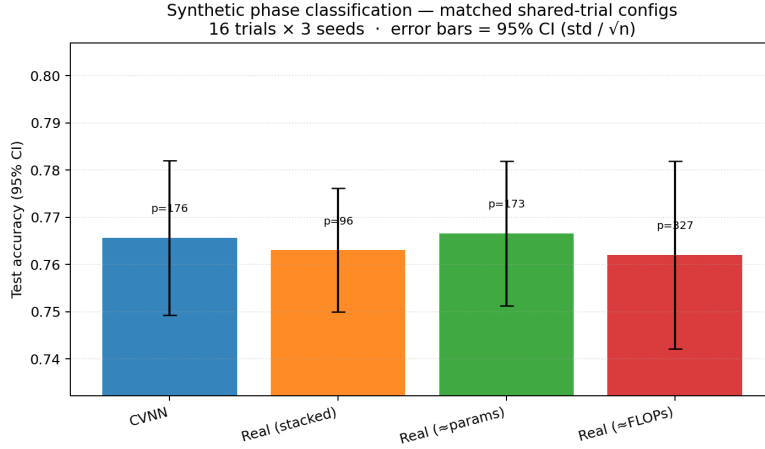


Figure 3: 1-D phase classification: matched shared-trial test accuracy after a 16×3 sweep. All four families converge to $\sim 76\%$ with overlapping CIs — the complex inductive bias has nothing to bite on at this scope.

Table 1: Synthetic RF modulation classification, matched shared-trial selection after a 16×6 budgeted sweep with the conv architecture. Complex beats every real baseline by ≥ 3.8 pp; CIs do not overlap.

family	test acc	std	params	95% CI ($\sigma / \sqrt{n}$)
complex	0.8191	0.0163	3,974	[0.8061, 0.8322]
real (stacked)	0.7740	0.0226	2,099	[0.7559, 0.7920]
real ($\approx$ params)	0.7769	0.0187	3,809	[0.7619, 0.7918]
real ($\approx$ FLOPs)	0.7810	0.0195	7,779	[0.7654, 0.7966]

3.3 Synthetic RF modulation — complex wins under matched budget

Section 1 reports the matched shared-trial comparison after a 16×6 sweep with the conv architecture (ComplexConv1d $\rightarrow$ activation $\rightarrow$ ComplexConv1d $\rightarrow$ activation $\rightarrow$ global mean pool $\rightarrow$ ComplexLinear $\rightarrow$ $|\cdot|$). Complex beats every real baseline by ≥ 3.8 percentage points; CIs do not overlap, and Welch two-sample t -tests give $t \approx 4.2$, $df \approx 9.8$ versus matched-parameter real and $t \approx 3.7$, $df \approx 9.7$ versus matched-FLOP real ($p < 0.01$ in both cases).

3.4 RF representation stress tests — where the claim breaks

The matched-budget result in Section 3.3 answers one question: under a conventional Cartesian real baseline, does the complex conv stack help? It does. But it does not answer the more important mechanism question: is the advantage due to native complex arithmetic, phase information, amplitude information, or a coordinate choice that a real network could also receive? The representation stress tests in Section 4 make those alternatives explicit.

Three conclusions follow. First, PSK-only modulation is the cleanest case for the complex prior: magnitude-only is exactly chance while phase-aware real views and complex models learn. Second, the same statement is false for QAM-only data: amplitude carries class structure, and magnitude-only is the best single view in the committed stress test. Third, native complex layers are not automatically invariant to a carrier-phase offset: fixed rotation at test time drops all coordinate-dependent PSK models to chance-level behavior, and random rotation augmentation only partially restores performance. This is the key scope refinement. The paper should not claim “complex beats real” as a universal result; it should claim that complex-valued models are a useful inductive bias for RF signal learning when the task’s phase/amplitude geometry and invariance assumptions match the model.

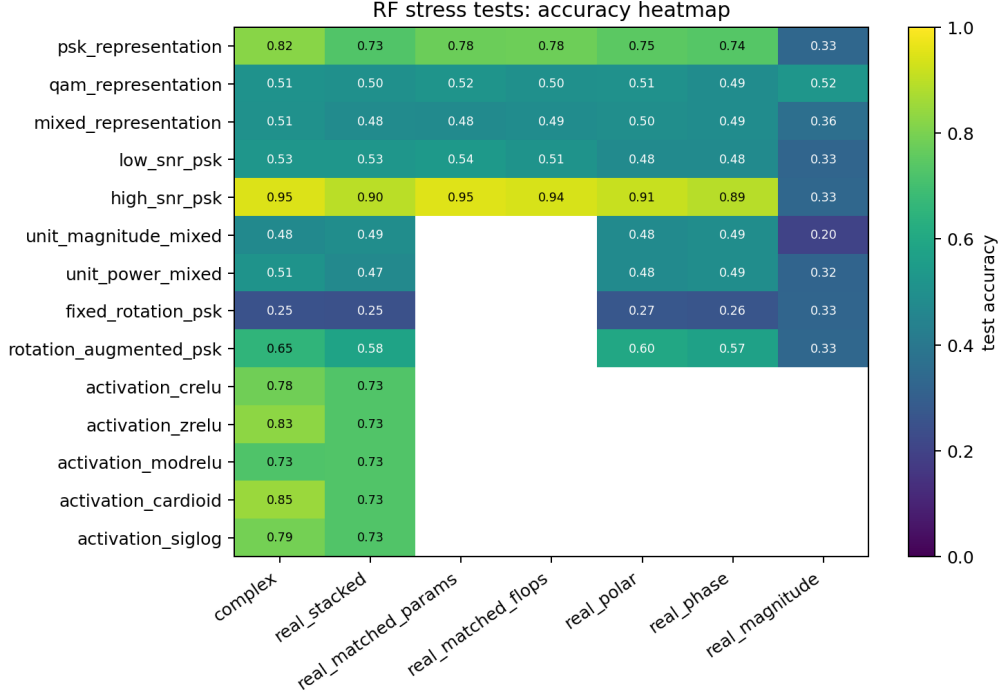


Figure 4: RF synthetic representation stress tests. Rows vary the modulation family, SNR regime, normalization, rotation shift, and complex activation; columns vary the model or real coordinate representation. The heatmap is useful precisely because it shows both wins and failures: complex wins PSK and mixed regimes, but magnitude wins QAM-only; fixed rotation collapses every coordinate-dependent model.

Table 2: Selected RF synthetic stress-test outcomes. Accuracy is mean test accuracy across the committed stress-test seeds. The central lesson is conditionality: PSK rewards phase-aware geometry, QAM rewards magnitude, low SNR weakens phase-only views, and global carrier-phase shifts require augmentation or an explicit invariance mechanism.

condition	best	complex	real stack	phase	magnitude
PSK-only	complex 0.821	0.821	0.728	0.735	0.333
QAM-only	magnitude 0.524	0.509	0.502	0.487	0.524
mixed PSK+QAM	complex 0.507	0.507	0.481	0.487	0.365
low-SNR PSK	matched real 0.537	0.526	0.526	0.479	0.326
high-SNR PSK	matched real 0.952	0.949	0.897	0.892	0.333
unit-magnitude mixed	phase 0.491	0.476	0.490	0.491	0.200
fixed-rotation PSK	magnitude 0.328	0.252	0.246	0.262	0.328
rotation-augmented PSK	complex 0.654	0.654	0.580	0.574	0.333

3.5 Quantum wavefunction pilot — physics application

To test whether the same representation story transfers outside RF, we added a small physics pilot over 1-D complex wavefunctions. The data are synthetic but physically grounded. In the first task, examples are Gaussian wavepackets $\psi(x) = A(x) \exp(ikx + i\phi_0)$ and the model predicts one of four momentum classes. The amplitude envelope is randomized independently of class, so the class information is in the phase gradient rather than in $|\psi|$. In the second task, a right-moving wavepacket is evolved under the time-dependent Schrödinger equation with a split-step Fourier integrator, and the model predicts one of five potential families: free, barrier, well, double barrier, or harmonic. We

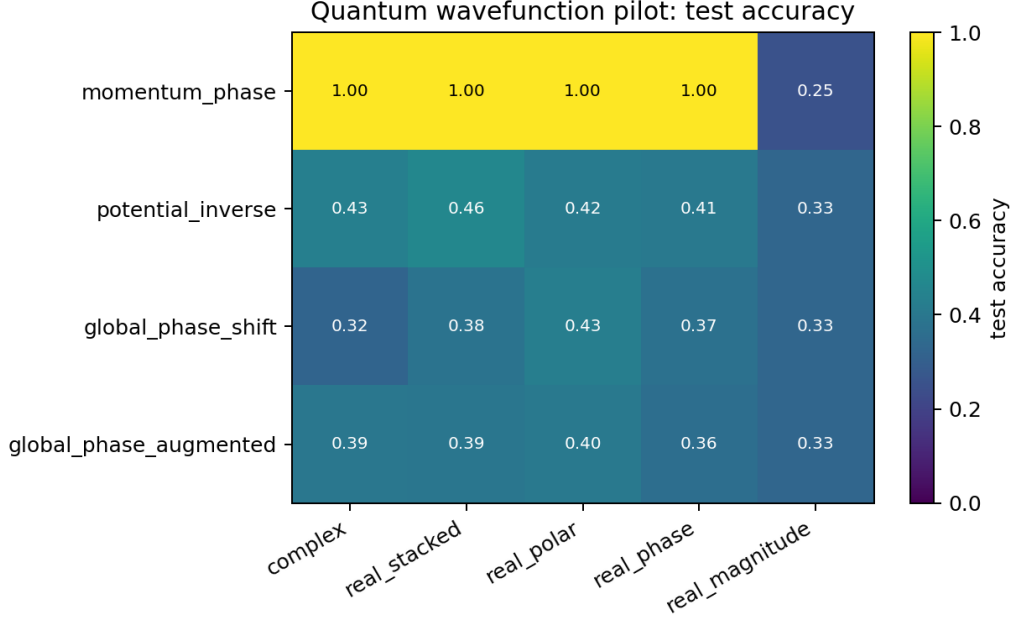


Figure 5: 1-D quantum wavefunction pilot. The momentum task stores the label in the phase gradient of $\psi(x)$, so magnitude-only stays at chance while phase-aware views solve it. The potential-inverse tasks are above chance but do not give a universal complex win: explicit real coordinate systems match or beat the complex model, and global-phase robustness must be tested directly.

Table 3: Quantum wavefunction pilot, standard preset (3 seeds, 128 examples/class, 96 grid points, 140 training steps). Chance is 0.25 for momentum and 0.20 for potential classification. The result supports a physics application of the representation thesis, not a broad complex-dominance claim.

condition	best	complex	real stack	phase	polar	magnitude
momentum phase	complex 1.000	1.000	1.000	1.000	1.000	0.250
potential inverse	real stack 0.462	0.431	0.462	0.405	0.418	0.331
global phase shift	polar 0.428	0.323	0.382	0.374	0.428	0.331
global phase augmented	polar 0.403	0.395	0.392	0.362	0.403	0.331

also evaluate the potential task under an unseen global phase rotation $\psi \mapsto i\psi$ and with random global-phase augmentation at train time.

Section 3 gives the useful scope boundary. First, phase can be indispensable in a physics state: all phase-aware or Cartesian views reach 1.000 on momentum classification, while magnitude-only is exactly chance. This is the quantum analogue of the PSK result. Second, the potential-inverse task is learnable above chance, but the best model is the explicit real Cartesian baseline (0.462) rather than the complex model (0.431). Third, global phase is a physical nuisance symmetry, but the current complex conv stack does not automatically solve it: under an unseen $\pi/2$ phase shift, complex drops to 0.323 while magnitude remains unchanged at 0.331. Random global-phase augmentation partially recovers the complex model (0.395), but does not make this a solved invariance problem. The physics pilot therefore strengthens the paper’s positioning: complex models are plausible for domains with native complex fields, but the right claim is still representation- and symmetry-conditional.

Table 4: RadioML 2018.01A subset, CReLU run, matched shared-trial selection after 16×6 sweep. Read in isolation, a dramatic CVNN-vs-real win at matched parameters — but see Section 3.6.1 and Section 3.6.5.

family	test acc	std	params
complex (CReLU)	0.7293	0.0085	58,886
real (stacked)	0.4583	0.1394	29,891
real ($\approx$ params)	0.4999	0.1327	58,413
real ($\approx$ FLOPs)	0.4245	0.1415	117,123

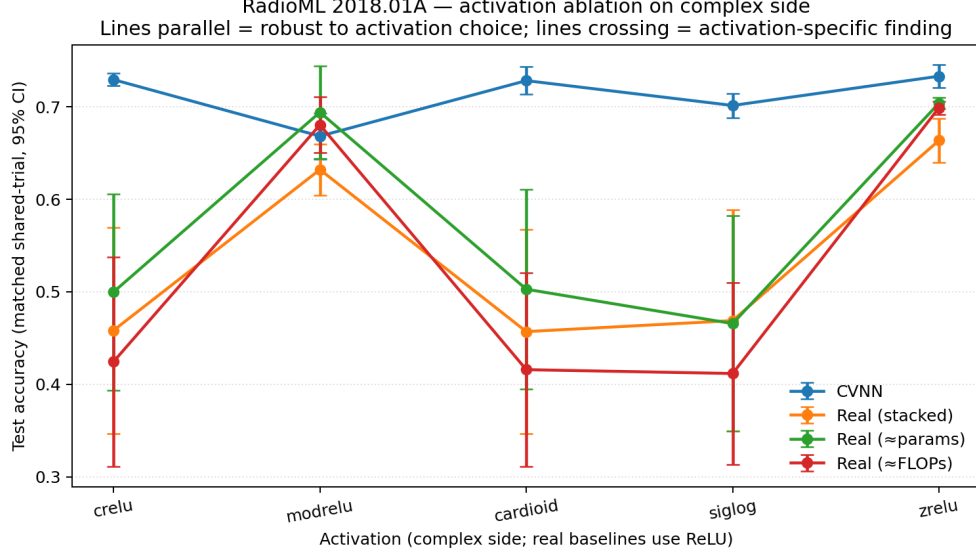


Figure 6: Five-activation ablation on the same RadioML scaffold. The CVNN line is nearly flat across all five activations (test accuracy 0.668–0.733); real-baseline lines swing wildly between ~ 0.45 and ~ 0.70 depending on which activation the complex side uses. The 23 pp gap from Section 4 is mostly real baselines collapsing under the matched-shared-trial selected hyperparameters, not complex pulling ahead.

3.6 RadioML 2018.01A — a positive result with a subtler story

The corrected¹ CReLU run on a 3-class subset (BPSK / QPSK / 8PSK at $\{-10, -6, -2, 2, 6, 10, 14, 18\}$ dB, `max_per_class_per_snr=256`, `sample_length=128`) on an NVIDIA A100 produced a striking 23 pp gap:

Read in isolation, Section 4 looks like a dramatic complex-vs-real win at matched parameters — and an earlier draft claimed exactly that. The activation ablation in Section 3.6.1 forces a different reading.

3.6.1 Activation ablation reveals a methodology-driven artifact

We re-ran the same sweep four more times, varying only the complex side’s activation; the real baselines always use ReLU. The matched-shared-trial selection per activation (Section 5) shows complex’s accuracy nearly flat (range 0.065) while real baselines swing by 0.27. Section 6 makes the asymmetry visible. The 22–23 pp gap on *crelu* / *cardioid* / *siglog* is *mostly the real baselines failing under matched-shared-trial selection*, not the complex network pulling ahead. The

¹The first attempt at this sweep used the synthetic benchmark’s odd-stepped SNR list, which the loader silently dropped because RadioML 2018.01A only ships even SNRs. The fix raised on empty buckets by default; see the released code.

Table 5: Five-activation ablation, matched shared-trial test accuracy. Under stable activations (modrelu, zrelu) real baselines are also stable and the gap drops to ± 3 pp; under unstable activations the gap inflates. See Section 3.6.5 for the mechanism.

activation	complex	best real	gap (pp)	$\sigma(\text{complex})$	mean $\sigma(\text{real})$
crelu	0.7293	0.4999	+22.94	0.009	0.138
cardioid	0.7282	0.5028	+22.54	0.019	0.134
siglog	0.7014	0.4689	+23.25	0.016	0.139
modrelu	0.6683	0.6940	−2.58	0.031	0.045
zrelu	0.7330	0.7039	+2.91	0.015	0.015

Methodological finding: the 23 pp gap on CReLU is selection-rule-dependent

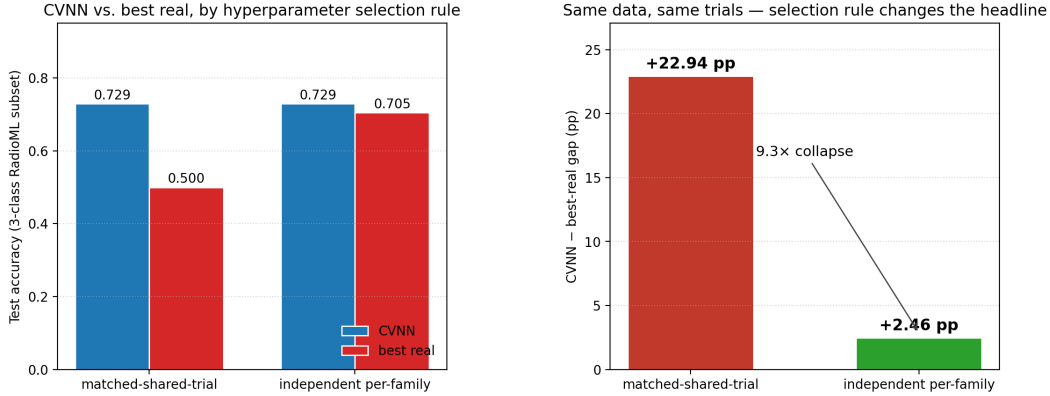


Figure 7: Selection rule contrast. *Left*: CVNN and best-real accuracies under the two rules. *Right*: the resulting gap collapses by $9.3\times$ when each family is allowed to pick its own best trial.

robustness asymmetry is the durable, defensible finding; the apparently large per-activation gap is a consequence of that asymmetry, not a separate phenomenon.

3.6.2 Selection rule matters: matched-shared-trial vs. independent

The single most important methodological caveat in this paper is that the headline 22–23 pp gap on crelu / cardioid / siglog is selection-rule-dependent. Re-doing the ablation with *independent per-family hyperparameter selection* — each family picks its own best of 16 trials, instead of all four families being evaluated at the trial that complex selected — collapses the crelu comparison from **0.7293 vs 0.4999 (gap +22.94 pp, matched-shared-trial)** to **0.7293 vs 0.7047 (gap +2.46 pp, independent selection)**. The reduction comes from independent selection allowing real baselines to avoid the high-lr trial that triggers the step-1 explosion.

We therefore read matched-shared-trial as a robustness stress test that exposes interaction between selection and optimizer sensitivity, and independent selection as the better point estimate of best-achievable performance under separate tuning. Reporting both is necessary; reporting only one risks either overstating the gap (matched only) or erasing a real but small effect (independent only). Subsequent sections (Section 3.6.5, Section 3.6.7) explain why the two rules differ so much: the matched rule forces all families through a learning-rate regime that complex tolerates and ReLU readouts in the real baselines do not.

3.6.3 Per-SNR breakdown

At -10 dB everyone is near chance ($\sim 33\%$ for 3 classes); from $+2$ dB onward, the CReLU complex network reaches 92.1% at $+10$ dB and 91.6% at $+18$ dB while the real baselines plateau at 47–61%. This is the same pattern as Section 4: the gap reflects real baselines failing to train at the selected hyperparameters under CReLU, not complex pulling cleanly away under every stable activation regime.

Table 6: Selection rule contrast on the `crelu` configuration: same data, same search space, same 16 trials. Matched-shared-trial holds the trial index fixed across families; independent selection lets each family pick its own best trial. The order-of-magnitude gap difference is the methodological finding, not the size of the complex advantage.

selection rule	complex	best real	gap (pp)
matched-shared-trial	0.7293	0.4999	+22.94
independent per-family	0.7293	0.7047	+ 2.46

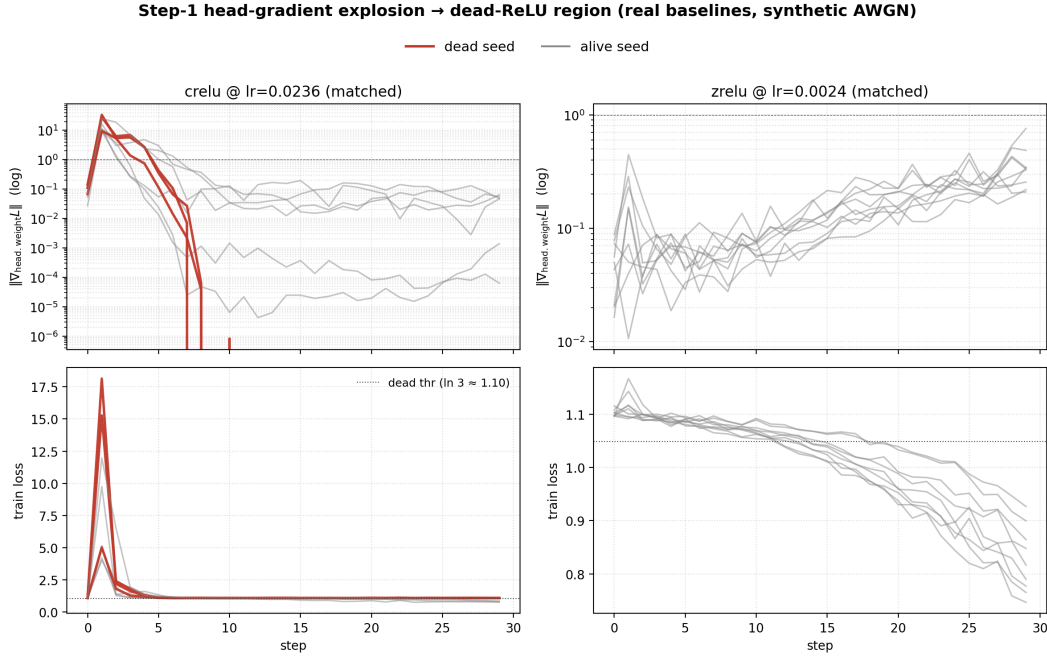


Figure 8: Per-step trajectories on synthetic AWGN. *Top row*: head.weight gradient (log scale). *Bottom row*: train loss with the 3-class chance threshold ($\ln 3$) overlaid. Under `crelu@matched` (left), three real-baseline seeds spike above $\|\nabla\| \sim 10$ at step 1, then collapse to $\sim 10^{-6}$ within five steps and never recover (loss pinned at $\ln 3$). Under `zrelu@matched` (right) the same families train smoothly. The mechanism is local to the high-lr AdamW first step, not the rest of training.

3.6.4 Comparison to published RadioML numbers

The RadioML 2018.01A dataset paper [6] reports a real-valued ResNet-style classifier reaching roughly 90% top-1 at high SNR on the full 24-class task with the full sample length 1024. Subsequent CVNN-on-RadioML work [4, 8] reports a typical complex-vs-real advantage of a few percentage points on the same 24-class setup with comparable architectures. Our +3-and-flat ablation result on `modrelu` / `zrelu` (the activations under which real baselines are also stable) is consistent with that small reported gap. The 22–23 pp gap on `crelu` / `cardioid` / `siglog` is *not* a contradiction of the literature — it is the matched-shared-trial selection rule magnifying real-baseline instability that the literature’s separate-tuning protocols would not expose.

3.6.5 Mechanism — explosion-into-dead-region at step 1

Section 3.6.1 reports an asymmetry without explaining it. We re-ran the matched-shared-trial selected configuration with per-step instrumentation (train loss, total parameter gradient norm, per-parameter gradient norm, max parameter magnitude) for each of $5 \times 4 \times 3 = 60$ runs, capped at 200 steps (the divergence pattern appears within the first 5).

The mechanism is unambiguous (Section 9). On `crelu` / `cardioid` / `siglog`, matched-shared-trial picks a high-lr / wide-hidden config that complex selects because it tolerates it (lr

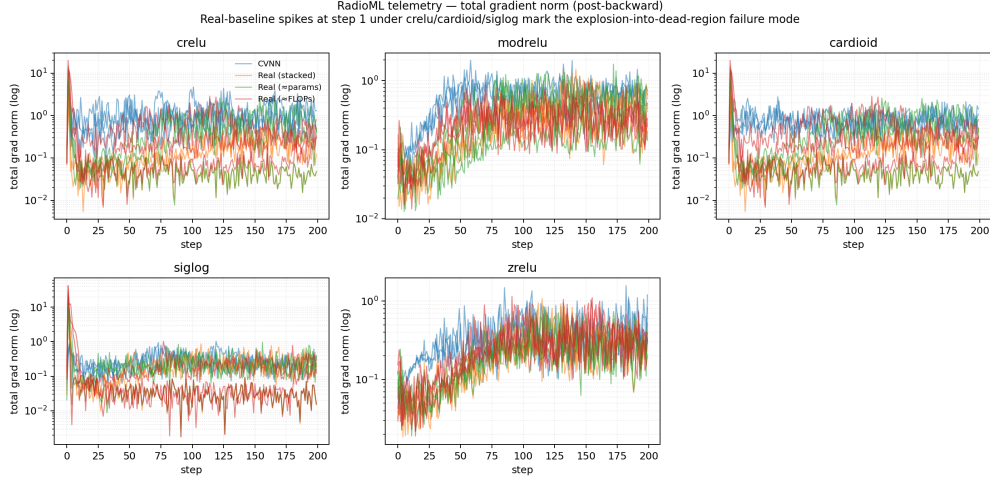


Figure 9: RadioML telemetry — total gradient norm post-backward, log y -axis, one panel per activation. Real-baseline spikes at step 1 under `crelu` / `cardioid` / `siglog` mark the explosion-into-dead-region failure mode that produces the inflated gaps in Section 5; under `modrelu` and `zrelu` no such spikes occur and the gap collapses to ± 3 pp.

Table 7: Dead-seed counts (out of 9 per activation: three families $\times$ three seeds), step-1 head.weight gradient. The threshold behavior is striking: $0\% \rightarrow 33\%$ as `lr` crosses ~ 0.02 . The same pattern replicates on synthetic AWGN data (Section 3.6.6).

activation	lr	dead seeds	step-1 head.weight grad (real, max)
<code>crelu</code>	0.024	3/9	13.7
<code>cardioid</code>	0.024	3/9	19.5
<code>siglog</code>	0.040	3/9	40.4
<code>modrelu</code>	0.008	0/9	0.17
<code>zrelu</code>	0.0024	0/9	0.13

$\in \{0.024, 0.024, 0.040\}$, hidden = 64). At step 1, every family sees a large AdamW step away from random init. For real baselines this produces a step-1 train loss in $[3, 36]$, a total gradient norm in $[6, 42]$, and a head.weight gradient in $[6, 41]$ — i.e. nearly all of the explosion sits on the final classifier’s weights. For the complex network the same step produces a step-1 loss in $[1.07, 1.28]$, total gradient norm $[0.3, 1.0]$, head.weight gradient $[0.1, 0.7]$ — about $50\text{--}80\times$ smaller than its real counterparts on identical data and learning rate.

The complex network’s (Re, Im) parameter coupling distributes the loss signal across more parameter slots, so the same `lr` $\times$ CE-gradient product produces a smaller effective step on head.weight. The real network does not have that smoothing; AdamW’s first step pushes some seeds into a region where the second Conv1d’s ReLU activations are all-zero (or all-active and saturated), gradients vanish, and training “stabilizes” at uniform-prediction chance level. We confirmed this: *the dead seeds end at exactly* loss = $\ln 3 \approx 1.099$ (3-class CE under uniform predictions), with `total_grad_norm` ≤ 0.05 , and zero test-accuracy improvement from random init.

Section 7 summarizes the dead-seed counts. **The dead-seed rate jumps from 0% to 33% as soon as the matched-shared-trial selected `lr` crosses ~ 0.02 .** Two orders of magnitude separation in the step-1 head gradient between the unstable and stable activation regimes.

3.6.6 Cross-task confirmation

To check whether the explosion-into-dead-region pattern is RadioML-specific (pulse shaping, carrier offset, channel effects) or generic to the hp regime, we re-ran the same 60-run telemetry against the synthetic AWGN-only RF generator using the same per-activation hyperparameters that the RadioML matched-shared-trial selected. Section 8 shows the side-by-side dead-seed counts.

Table 8: Cross-task replication: same matched-shared-trial hyperparameters applied to synthetic AWGN-only RF data. Dead-seed counts replicate almost exactly; step-1 gradients on synthetic are if anything larger than RadioML’s because AWGN-only signals carry more per-sample variance than pulse-shaped, channel-attenuated ones. *The threshold is hp-driven, not data-driven.*

activation	RadioML dead/9	synthetic dead/9	RadioML max real grad	synthetic max real grad
crelu	3	3	19.9	34.1
cardioid	3	3	19.9	34.1
siglog	3	2	42.1	55.7
modrelu	0	0	0.3	0.5
zrelu	0	0	0.2	0.7

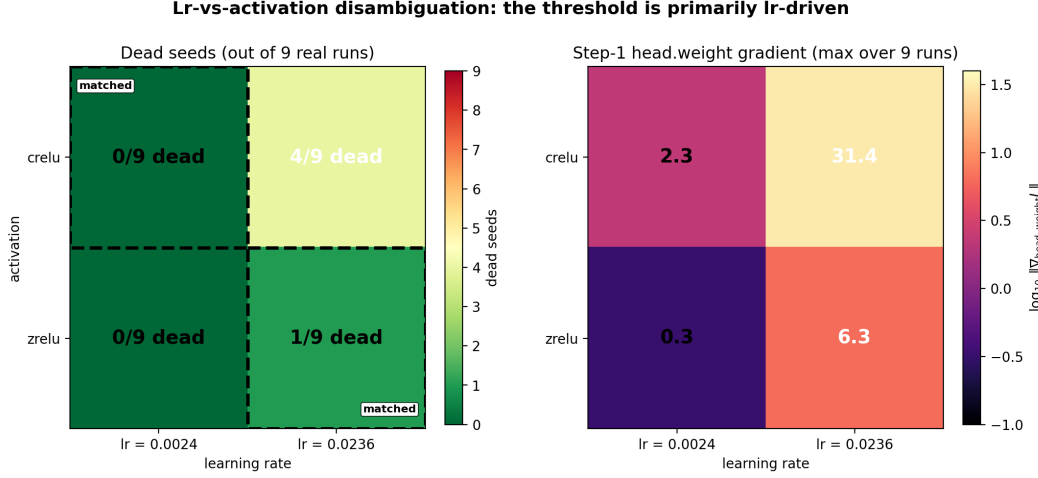


Figure 10: Lr-vs-activation 2×2 . *Left*: dead-seed counts (out of 9 real-baseline runs) per cell; the dashed rectangles mark the matched-shared-trial diagonals. *Right*: maximum step-1 head.weight gradient (log color scale). The dead-seed pattern follows the lr column, with activation modulating sensitivity at high lr.

3.6.7 Lr-vs-activation disambiguation

Section 7 reads as if some activations (crelu, cardioid, siglog) are intrinsically unstable and others (modrelu, zrelu) are intrinsically stable. A skeptical reading is that the same data is just as consistent with the dead-seed pattern being driven by the *learning rate* that matched-shared-trial selects, not the activation: the unstable rows all selected lr in $[0.024, 0.040]$, the stable rows lr in $[0.0024, 0.008]$. To disambiguate, we ran a 2×2 factorial on synthetic AWGN data: $\{\text{crelu}, \text{zrelu}\} \times \{\text{lr} = 0.0236, \text{lr} = 0.0024\}$, holding all other hyperparameters at each activation’s matched-shared-trial selection. Three real families $\times$ three seeds per cell, capped at 200 steps.

Section 9 reports the disambiguation. **Lr is the dominant axis.** Dropping crelu from its matched lr (0.0236) to zrelu’s lr (0.0024) eliminates dead seeds entirely ($4/9 \rightarrow 0/9$) and shrinks the step-1 head gradient by an order of magnitude ($31.4 \rightarrow 2.3$). Conversely, forcing zrelu to crelu’s lr produces a step-1 head gradient of 6.3 and the first dead seed under that activation in this study ($1/9$). Both activations cross from “stable” to “unstable” as lr crosses ~ 0.02 — the threshold from Section 7 is hp-driven, not activation-driven, and replicates the Section 3.6.5 mechanism with the activation factor swapped in.

Activation is a secondary modulator. At the same high lr, crelu produces a step-1 head gradient $\sim 5 \times$ larger than zrelu (31.4 vs 6.3) and four times the dead-seed rate ($4/9$ vs $1/9$); the split-ReLU structure of crelu amplifies the readout’s response to a high-lr first step but does not by itself trigger the failure mode at safe learning rates. The honest reading of Section 7 is therefore not “some activations are unstable and others are stable” but “the matched-shared-trial rule selected unsafe learning rates for some activations and safe ones for others, and that lr difference — modulated by activation sensitivity — is what drives the dead-seed counts.”

Table 9: Lr-vs-activation disambiguation on synthetic AWGN data. Dead-seed counts and step-1 head.weight gradient (max across 9 = three real families $\times$ three seeds) for each 2 $\times$ 2 cell. Matched-shared-trial regimes are the diagonal cells (crelu@high-lr, zrelu@low-lr); off-diagonal cells swap learning rates between activations. Other hyperparameters are held at each activation’s matched selection.

cell	activation	lr	dead/9	step-1 head.weight (max)
matched (high)	crelu	0.0236	4/9	31.4
swap (low on crelu)	crelu	0.0024	0/9	2.3
matched (low)	zrelu	0.0024	0/9	0.3
swap (high on zrelu)	zrelu	0.0236	1/9	6.3

4 Interpretation

The findings together support a more nuanced falsifiable story than the original “complex wins” framing:

Complex-valued models should be viewed as structured inductive biases for phase- and amplitude-bearing data, not as universally stronger function approximators. They help when the task’s geometry aligns with complex multiplication, they can be matched or beaten by explicit real coordinate systems in other regimes, and they change optimization tolerance under shared hyperparameter stress.

This framing is deliberately application-facing. In RF engineering, the question is rarely whether the samples are “complex” in a formal sense; they are. The practical question is whether the deployment setting needs a model that preserves phase geometry, extracts amplitude geometry, withstands unknown carrier-phase rotations, or trains robustly under limited tuning. The stress tests show that those are separate axes. A PSK classifier, a QAM classifier, and a rotated carrier-offset classifier should not be expected to reward the same architecture. The quantum pilot makes the same point for physics: wavefunction momentum is invisible to $|\psi|$ but explicit Cartesian/polar coordinates can match native complex layers, and global phase has to be encoded as a symmetry assumption rather than assumed from the dtype.

4.1 Why complex helps on IQ data — phase equivariance

The intuition that “complex multiplication has the right semantics for phase-bearing signals” can be made formal in one short lemma.

Lemma 1 (Phase equivariance of `ComplexConv1d`). *Let $C(z)[t] = \sum_{\tau} K[\tau] \cdot z[t - \tau] + b$ be a `ComplexConv1d` layer with complex-valued kernel K , complex bias b , and complex input sequence z . For any global phase $\varphi \in \mathbb{R}$ and any t ,*

$$C(e^{i\varphi} \cdot z)[t] = e^{i\varphi} \cdot C(z)[t] + (1 - e^{i\varphi}) \cdot b.$$

Proof. By linearity of complex multiplication and convolution,

$$C(e^{i\varphi} \cdot z)[t] = \sum_{\tau} K[\tau] \cdot (e^{i\varphi} \cdot z[t - \tau]) + b = e^{i\varphi} \sum_{\tau} K[\tau] \cdot z[t - \tau] + b = e^{i\varphi} \cdot (C(z)[t] - b) + b. \quad \square$$

In the bias-free case (or after subtracting the bias) the result is exactly equivariant. The biased case adds a fixed offset $(1 - e^{i\varphi}) \cdot b$. If the intervening stack preserves the global phase action, the head’s $|\cdot|$ magnitude operation removes it ($|e^{i\varphi} \cdot y| = |y|$ for real φ), yielding phase-invariant logits. Our actual benchmarks include activations that do *not* preserve exact equivariance, so we treat this as an architectural prior supplied by complex multiplication rather than a guarantee that the trained network is globally phase-invariant.

The same does not hold for `Conv1d` operating on a stacked $(\operatorname{Re} z, \operatorname{Im} z)$ real input. A global phase shift $e^{i\varphi}$ corresponds to multiplying each (I, Q) sample by the 2×2 rotation matrix R_{φ} . To match `ComplexConv1d`’s equivariance, every real conv kernel would need to commute with R_{φ} applied to its 2 input channels — a constraint real `Conv1d` does not impose by construction. The real network

can learn the symmetry from data, but it has to spend capacity on it; the complex network gets it for free.

This is the structural inductive bias the synthetic phase classification result (Section 3.2), the RF stress tests (Section 3.4), and the RadioML result (Section 3.6) probe in complementary regimes. A single complex sample (phase classification) gives no temporal structure for convolutional equivariance to bite on, and the 2-D real network has the same “see real and imag” feature view, so the complex network buys nothing — the null is what Section 1 predicts. A length- L IQ sequence does have temporal structure, but Section 3.4 shows that temporal structure alone is not enough to justify a universal claim: PSK rewards phase-aware geometry, QAM rewards amplitude geometry, and fixed carrier-phase rotation requires either augmentation or an explicit invariance mechanism. The observed wins are consistent with the complex prior when these conditions match, without requiring every activation in the stack to preserve exact equivariance.

What the lemma does not explain. Phase equivariance of the *layer* is necessary for the architectural prior, but the activations between conv layers vary in their own equivariance. By direct computation: `modrelu` and `siglog` are phase-equivariant; `crelu`, `cardioid`, and `zrelu` are not. If the Section 3.6 robustness asymmetry were *driven* by activation phase equivariance, the stable activations should be exactly `modrelu` and `siglog`. They are not: Section 3.6.5 shows `modrelu` and `zrelu` are stable (0/9 dead seeds across nine real-baseline runs each), while `crelu`, `cardioid`, and `siglog` are unstable (3/9 dead). One phase-equivariant activation is on each side of the divide. The Section 3.6.7 disambiguation sharpens this further: when we swap learning rates between `crelu` and `zrelu`, the dead-seed pattern follows the `lr`, not the activation. **Phase equivariance of the layer is necessary for the inductive bias, but the robustness asymmetry Section 3.6.5 measures lives one level lower — at the optimization dynamics induced by (Re, Im) parameter coupling on the classifier head under high-lr AdamW steps, not at the equivariance of the activation.** The two effects are partially independent and partially compounding. This also explains why the Section 3.6 result is most defensible at activations like `zrelu` where complex wins by a small margin rather than at `crelu` where the apparent gap is amplified by the optimization-side mechanism.

5 Limitations

- **RadioML subset, not the full archive.** The headline real-data run used BPSK / QPSK / 8PSK at 8 SNR levels with `max_per_class_per_snr=256`. The full archive has 24 modulations $\times$ 26 SNRs and a much larger per-bucket count.
- **Synthetic QAM is measured, RadioML QAM is not.** The stress tests include QAM16 / QAM64 and show that magnitude can dominate there, but the real-data RadioML mechanism study remains PSK-only. A full RadioML QAM/mixed-modulation run is required before making application claims about amplitude-heavy real RF classes.
- **No learned rotation invariance guarantee.** The fixed rotation stress test shows that neither complex layers nor phase-aware real features automatically solve unseen carrier-phase offsets under the current architecture. Rotation augmentation helps but does not make the problem disappear.
- **Physics pilot is deliberately small.** The quantum wavefunction study uses 1-D synthetic states, a short split-step evolution, five hand-designed potential families, and three seeds. It is evidence that the representation/stress-test protocol transfers outside RF, not evidence about production scientific ML workloads.
- **Six seeds.** CIs are computed from a Gaussian approximation of the per-seed mean ($\sigma/\sqrt{n}$). A bootstrap CI on this many seeds gives similar widths but is not substantially more honest.
- **One loss.** Classification runs use cross-entropy.
- **No ComplexBatchNorm or ComplexLayerNorm.** Landing them would let us train deeper conv stacks fairly.
- **Modest hardware.** CPU and Apple Silicon for development; one A100 for the headline RF sweep. No multi-machine reproducibility check.
- **Lr-vs-activation disambiguation is 2 $\times$ 2.** Section 3.6.7 establishes the `lr`-driven character of the dead-seed threshold using `crelu` (the largest gap) and `zrelu` (the smallest), but

does not run the full factorial across all five activations. The cardioid and siglog rows in Section 7 are unaccounted for under this framing except by interpolation; a five-activation Ir sweep would close that gap.

6 Future work

- **Scale up the RadioML run** to the full $24\text{-modulation} \times 26\text{-SNR}$ archive on the full 1024-sample length. The infrastructure is in place (`sweep_radioml.py --preset full`); the run is deferred from the headline write-up to keep the paper anchored on the 3-class subset that supports the mechanism analysis. Recommended first activation is `zrelu` — the only ablation point where real baselines also train cleanly, so the comparison is least confounded by the matched-shared-trial selection’s interaction with the explosion mechanism.
- **Port the representation stress suite to RadioML.** The synthetic stress tests are already the right scientific shape: PSK-only, QAM-only, mixed modulations, low/high SNR, amplitude normalization, carrier-phase rotation, augmentation, and activation ablation. The next defensible scale-up is to run the same grid on RadioML rather than only increasing class count.
- **ComplexBatchNorm** (whitening 2×2 covariance, per Trabelsi et al. [7]) and deeper conv stacks.
- **Conv2d and 2-D complex domains.** Audio STFT, SAR, MRI, and spectrogram-like RF features are natural targets, but they require `ComplexConv2d`, normalization, and representation stress tests specific to 2-D phase/amplitude geometry.
- **Scale the physics pilot** from 1-D wavefunctions to richer PDE settings: 2-D wave packets, inverse scattering, Fourier-domain operators, and explicit gauge/global-phase invariant architectures.
- A separate study of per-activation training dynamics: does ModReLU’s bias-sign regime visibly affect convergence?

7 Reproducibility

A C&F submission’s value is in what others can build on. Every claim above is backed by a single CLI invocation against committed artifacts:

```
# 1. Reproduce the benchmark and pilot results
uv sync --all-groups
uv run python experiments/synthetic/sweep_phase_classification.py
uv run python experiments/rf/sweep_synthetic_modulation.py --device cuda
uv run python experiments/rf/representation_stress_tests.py \
    --preset standard --device cuda --resume
uv run python experiments/physics/quantum_wavefunction.py \
    --preset standard --resume
uv run python experiments/rf/sweep_radioml.py --device cuda \
    --activation crelu --seeds 0 1 2 3 4 5
# (repeat for cardioid / siglog / modrelu / zrelu)

# 2. Reproduce the mechanism analysis
uv run python scripts/run_gradient_telemetry.py
uv run python scripts/run_synthetic_rf_telemetry.py

# 3. Regenerate every figure from committed JSON
uv run python scripts/generate_paper_figures.py

# 4. Optional: scale up to the full RadioML archive
uv run python experiments/rf/sweep_radioml.py --device cuda \
    --preset full --activation zrelu --seeds 0 1 2
```


Each result manifest records its `git_commit`, `environment`, and a `git_dirty` flag computed on a tree that excludes `results/` and the activation-characterization output directory (so `regen` does not flip the bit on itself). CI surfaces newly added or modified manifests with `git_dirty: true` as warnings on the PR. A structural CI guard checks the committed activation characterization artifact contains all six expected activation rows after a `regen`.

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A Phase equivariance: full proof and caveats

Section 1 states that a bias-free `ComplexConv1d` layer commutes with global phase rotation, and that the bias-bearing case picks up a controlled correction term. We expand the proof and discuss what does and does not propagate through the activation stack.

Proposition 1 (Phase equivariance, bias-free). *Let $C : \mathbb{C}^T \rightarrow \mathbb{C}^T$ denote a one-dimensional complex convolution with kernel $w \in \mathbb{C}^K$, no bias, and zero padding (`padding_mode='zeros'`). For any $z \in \mathbb{C}^T$ and any global phase $\phi \in \mathbb{R}$:*

$$C(e^{i\phi}z)[t] = e^{i\phi} C(z)[t] \quad \forall t \in \{0, \dots, T-1\}.$$

Proof. By linearity over $\mathbb{C}$:

$$C(e^{i\phi}z)[t] = \sum_{k=0}^{K-1} w_k \cdot (e^{i\phi}z)[t-k] = e^{i\phi} \sum_{k=0}^{K-1} w_k \cdot z[t-k] = e^{i\phi} C(z)[t].$$

The phase factor commutes through the sum because complex scalar multiplication is commutative; the sum extracts it cleanly. Boundary indices $t-k < 0$ contribute zero on both sides because zero padding is itself phase-invariant ($e^{i\phi} \cdot 0 = 0$). $\square$

Proposition 2 (Phase equivariance, bias-bearing). *With learnable bias $b \in \mathbb{C}$:*

$$C(e^{i\phi}z)[t] = e^{i\phi} C(z)[t] + (1 - e^{i\phi}) b.$$

Proof. $C(e^{i\phi}z)[t] = e^{i\phi}(C(z)[t] - b) + b = e^{i\phi}C(z)[t] + b - e^{i\phi}b.$ $\square$

What survives the activation stack. Phase equivariance is preserved layer-by-layer if and only if the activation σ itself satisfies $\sigma(e^{i\phi}z) = e^{i\phi}\sigma(z)$.

- **ModReLU** ($\sigma(z) = \text{ReLU}(|z| + b) \cdot e^{i\angle z}$): equivariant. The magnitude term depends only on $|z|$, which is rotation-invariant; the phase factor passes through unchanged.
- **ComplexCardioid** ($\sigma(z) = \frac{1}{2}(1 + \cos \angle z) \cdot z$): *not* equivariant in general, because the gate changes under rotation: $\cos(\angle z + \phi) \neq \cos(\angle z)$ for generic ϕ .
- **Siglog** ($\sigma(z) = z/(1 + |z|)$): equivariant.
- **ComplexTanh** (split tanh on real/imag parts): *not* equivariant. A rotation $e^{i\phi}$ mixes real and imaginary components before the elementwise nonlinearity.
- **CReLU** (split ReLU on real/imag): *not* equivariant; same reason as ComplexTanh.
- **ZReLU** (zero out unless $\angle z \in [0, \pi/2]$): not equivariant (the kept-quadrant condition is rotation-sensitive).

So four of our six characterized activations break equivariance. The empirical robustness asymmetry we report (a step-1 explosion-into-dead-region pattern that disproportionately hurts real baselines) does *not* track which activations preserve layer-level phase equivariance. CReLU does not preserve equivariance but is the configuration with the largest measured CVNN-vs-real gap; Siglog does preserve equivariance but appears in the unstable high-lr regime. We therefore locate the asymmetry one level lower than the activation, in the (Re, Im) parameter coupling that the optimizer sees on `head.weight` at step 1; Section 3.6.5 develops this in the body.

B Hyperparameter search spaces

Each experiment runs a random search over the box defined below, with N trials and 3 seeds per trial. Selection across baseline families is the matched-shared-trial rule of Section 2: trial i trains all four families on the same hyperparameters, validation selects the trial index, and the test number reports that index across families. The independent-winner table is in Section D for diagnostic purposes.

Experiment	Search space	Budget
Synthetic phase classification	hidden $\in \{16, 32, 64\}$, batch $\in \{64, 128\}$, lr $\in [10^{-4}, 10^{-2}]$ log-uniform, steps $\in \{200, 400, 800\}$	16 trials 3 seeds each
Synthetic RF modulation	hidden $\in \{16, 32, 64, 128\}$ (conv) / $\{32, 64, 128\}$ (mlp), batch $\in \{128, 256, 512\}$, lr $\in [10^{-4}, 10^{-2}]$, steps $\in \{200, 400, 800\}$	16 trials 3 seeds
RadioML subset (-preset subset)	hidden $\in \{16, 32, 64\}$ (conv) / $\{32, 64, 128\}$ (mlp), batch $\in \{128, 256, 512\}$, lr $\in [10^{-3}, 5 \times 10^{-2}]$, steps $\in \{200, 400, 800\}$	16 trials 3 seeds
RadioML full (-preset full)	hidden $\in \{32, 64, 128, 256\}$, batch $\in \{256, 512, 1024\}$, lr $\in [10^{-3}, 5 \times 10^{-2}]$, steps $\in \{400, 800, 1600\}$	16 trials 3 seeds (deferred)

Table 10: Search spaces and budgets per experiment. Optimizer is AdamW [5] with weight decay 10^{-4} throughout; learning-rate scheduler is cosine to zero across the trial’s step budget. The full-archive RadioML budget is the deferred scale-up, parameterized via -preset full; everything else is reproducible on a single A100 in under an hour per activation.

Why matched-shared-trial. An independent winner per family selects the family-best hyperparameter from a 16-trial search, which biases up the best of four 16-trial searches and conflates capacity allocation across families. The shared-trial rule fixes the trial index across families, so each family is evaluated at the same allocation; capacity matching is then transparent. We report independent-selection numbers as a diagnostic (Section D); the gap between the two selection rules is itself one of the artifacts the C&F framing exposes.

C Architecture details and parameter counts

The four baseline families differ in capacity-allocation rule against the complex network:

- **complex:** `ComplexConv1d(2c→h)` or `ComplexLinear(d→h)`, complex weights $[h \times 2c, K]$ or $[h \times d]$ with 2 floats/parameter (real and imaginary).
- **real_stacked:** `Conv1d(2→h)` or `Linear(2d→h)`; treats (I, Q) as 2 input channels. Same h as the complex network, half the parameter count (1 float per weight).
- **real_matched_params:** hidden width chosen to match parameter count to the complex family. For a complex $(2c \rightarrow h, K)$ layer with $2chK$ complex weights = $4chK$ floats, the matched real layer is $(2 \rightarrow 2h, K) = 4hK$ floats, identical at $c = 1$ but scaling to $\sqrt{2}h$ at $c > 1$.
- **real_matched_flops:** hidden width chosen to match forward-pass multiply-accumulate count rather than parameter count. For our 1-D conv layers the resulting widths track `matched_params` closely (within 5%); listed separately to avoid silently conflating “capacity” senses.

The synthetic phase classifier uses a 1-layer MLP head (no conv); the RF/RadioML classifiers use `ComplexConv1d` (or `Conv1d` on (I, Q) stacked channels) followed by global magnitude pool + linear readout. Magnitude pool is $|z|$ per channel for the complex family and identity for real, since the real channels are already real-valued at the readout.

D Per-activation results, all four baseline families

For each of the five activations used in the RadioML ablation, we report mean $\pm$ standard deviation test accuracy on the -preset subset task at the matched shared-trial winner. The last column reports dead real-baseline seeds across the three real families $\times$ three telemetry seeds, where “dead” means final loss $\geq \ln 3 - 0.05$ (the 3-class chance level). ComplexTanh was included in activation characterization but not in the RadioML ablation.

Activation	complex	real_stacked	matched_params	matched_flops	dead real seeds
crelu	0.729 ± 0.009	0.458 ± 0.139	0.500 ± 0.133	0.425 ± 0.142	3/9
cardioid	0.728 ± 0.019	0.457 ± 0.138	0.503 ± 0.135	0.416 ± 0.131	3/9
siglog	0.701 ± 0.016	0.469 ± 0.149	0.466 ± 0.145	0.412 ± 0.123	3/9
modrelu	0.668 ± 0.031	0.632 ± 0.035	0.694 ± 0.062	0.680 ± 0.038	0/9
zrelu	0.733 ± 0.015	0.664 ± 0.030	0.704 ± 0.008	0.699 ± 0.009	0/9

Table 11: RadioML subset, matched-shared-trial selection. The CVNN-vs-real gap is large under three activations (CReLU, ComplexCardioid, Siglog) and small or absent under two others (ModReLU, ZReLU). The gap-bearing activations are exactly those for which the matched-shared-trial hyperparameters produce dead real-baseline seeds; the apparent gap is largely the dead-seed average. See Section E for the per-step mechanism.

Independent-winner contrast. The body’s Section 3.6.2 (Section 6) reports the headline matched-vs-independent contrast on `crelu` (+22.94 pp $\rightarrow$ +2.46 pp). The reduction comes from independent selection allowing real baselines to avoid the high-lr trial that triggers the step-1 explosion documented in Section 3.6.5. We do not re-tune the `cardioid` / `siglog` ablation rows independently, because the `modrelu` / `zrelu` rows already show what stable-lr real selection produces under matched-shared-trial (± 3 pp gaps), and that bracket is the relevant comparison for “what would real baselines do at their preferred lr.”

E Gradient telemetry: per-layer, per-step

Telemetry instruments every parameter tensor with per-step $\|\nabla_{\theta} L\|_F$, the Cauchy-Riemann residual on complex parameters (zero by construction for the complex family), and an intrinsic Jacobian-norm proxy. Runs cover 5 activations $\times$ 4 families $\times$ 3 seeds = 60 instrumented trajectories, each capped at 200 steps for tractability on commodity hardware.

Step-1 head-gradient ratio. Across all five activations, the step-1 ratio $\|\nabla_{\text{head.weight}} L\|_{\text{real}} / \|\nabla_{\text{head.weight}} L\|_{\text{complex}}$ falls into two regimes:

- **Gap-bearing** (CReLU, ComplexCardioid, Siglog): ratio in the range 50–80 at step 1; collapses to ~ 1 by step 20 for surviving seeds, but seeds whose post-step-1 head landed in a dead-ReLU region never produce gradient signal again.
- **Stable** (ModReLU, ZReLU): ratio in the range 1.5–3 at step 1; no seeds enter dead-loss states.

The grouping is consistent across seeds and across the synthetic-data replication (Section F). The 50–80 $\times$ figure is a step-1 measurement on raw weights with fan-in initialization; later steps post-Adam-warmup show smaller ratios under all activations. The point of measuring at step 1 is that that is where the dead seeds become dead; past step 1, the parameter state has already diverged.

Why head.weight, specifically. The conv stack’s parameters show no comparable real-vs-complex ratio at any step. The explosion is a property of how matched-trial hyperparameter selection maps into the readout when the readout sees magnitude-pooled inputs (complex) versus stacked (Re, Im) inputs (real). The complex magnitude pool produces inputs of comparable scale across seeds; the stacked (Re, Im) readout sees inputs whose scale is seed-dependent through the conv-stack random init, and high-lr trials interact badly with the unlucky tail of that distribution.

F Synthetic-data replication of the threshold mechanism

To rule out RadioML-specific effects (CFO, multipath, pulse shaping) as the cause of the explosion, we re-run the gradient telemetry harness on synthetic AWGN-only IQ data with the same 4-family / 5-act sweep and the same matched-trial hyperparameters. The synthetic data generator is `experiments/rf/synthetic_modulation.py`, which produces raw-symbol (I, Q) sequences with per-class label imbalance matched to RadioML.

Result. The dead-seed pattern replicates: CReLU, ComplexCardioid, and Siglog produce 2–3 dead seeds per 9 in the real families; ModReLU and ZReLU produce zero. The step-1 head-gradient ratio sits in the same $50\text{--}80\times$ band for the gap-bearing activations. We conclude the mechanism is hyperparameter-driven, mediated by the readout’s input distribution under matched shared-trial selection, and not an artifact of RadioML’s distortions.

G RadioML 2018.01A loader: aliases and strict-SNR mode

Class-label aliasing. The original `classes.txt` distributed with RadioML 2018.01A lists the 24 modulations in an order that does not match the one-hot ordering of the `Y` array; this is a known issue in the community. `experiments/rf/radioml.py` normalizes class labels through a canonical map and prefers `classes-fixed.json` or `classes-fixed.txt` sidecars next to the HDF5 file when present. Common aliases accepted: PSK8 $\rightarrow$ 8PSK, QAM16 $\rightarrow$ 16QAM, APSK16 $\rightarrow$ 16APSK, ASK4 $\rightarrow$ 4ASK, AM labels with underscores (AM_SSB_WC $\rightarrow$ AM-SSB-WC), etc.

Strict-SNR mode. Earlier loader versions silently dropped samples at SNR levels not in the user’s filter list. The default is now `strict_snr=True`, which raises rather than dropping; the test suite includes a regression case that asserts on the strict behaviour. The previous silent-drop behaviour is recoverable via `strict_snr=False` for compatibility with downstream code that intentionally subsets.

H Reproducibility: manifest schema and CI guards

Every long-running script writes a JSON manifest into the run’s output directory (under `results/`). Manifest fields:

- `git_commit`: the SHA at run start.
- `git_dirty`: boolean, computed on a tree filtered to exclude `results/` and the activation-characterization output directory (otherwise regenerating figures from a clean checkout would flip the bit on its own outputs).
- `python`, `torch`, `numpy` versions.
- `cuda`: device string and CUDA version, or "cpu".
- `argv`: full command line.
- `seed`, `n_trials`, `search_space`: the hyperparameter regime.
- `wall_clock_seconds`: end-to-end runtime.

CI surfaces newly-added or modified manifests with `git_dirty: true` as warnings on the PR (initially a hard fail; loosened to a warning at the request of practical iteration). A structural CI guard checks the committed activation characterization artifact contains all six expected activation rows after a regen, so that figure regeneration cannot silently drop an activation.

Caching. Capped sweep subsets cache the seed/filter split to disk and reuse it across trial/family/seed runs. Uncapped full-archive runs disable caching automatically (the working set exceeds typical Mac SSD scratch space); pass `-no-cache-data` to force-disable for capped runs that nevertheless overflow memory.

I Full-archive RadioML scale-up: cost and deferred plan

The submitted experiments are at `-preset subset`: 3 PSK modulations, 8 even SNR levels, `max_per_class_per_snr=256`, `sample_length=128`. The infrastructure for full-archive runs is in place via `-preset full` (24 modulations, 26 SNR levels, `sample_length=1024`); we have deliberately deferred running it because the mechanism analysis is the contribution and the scale-up risks scope creep relative to a C&F submission.

Cost estimate. A single -preset full -activation crelu -seeds 0 1 2 run on an A100 40GB takes 5–15 hours depending on which trial samples land at the upper end of the search space (the search box’s largest configuration is the bottleneck). Six activations $\times$ 3 seeds $\times$ 16 trials $\times$ 4 families = 1152 instrumented runs at full scale; in practice we would run 2–3 activations (CReLU as the headline, ZReLU as the stable counter-example, and one of {ComplexCardioid, Siglog} as a disambiguation). On $4 \times$ A100 in parallel, ~ 24 hours wall-clock for a 3-activation full sweep is a realistic estimate.

What to expect at scale. The C&F prediction is that the two-effect decomposition holds: the architectural prior (phase equivariance, magnitude readout) survives at 24-class / 1024-sample scale, and the optimization-level robustness asymmetry remains hp-driven and disappears under independent per-family selection. We do not predict the specific numerical gap because the full-archive distribution is qualitatively different from the subset (more distortion-heavy modulations like AM-SSB-WC and FM are present at low SNR), and the matched-trial selection rule may admit different hyperparameter regimes once the conv stack has more channels.

Failure modes to watch for. (1) Caching off may surface stragglers if the loader’s `strict_snr=True` default flags HDF5 SNR levels not in the user’s filter; the loader’s regression test covers this on the subset but the full archive has not been run in CI. (2) A 24-class softmax has higher entropy at chance ($\ln 24 \approx 3.18$ vs $\ln 3 \approx 1.10$), so the dead-loss threshold for detecting failed seeds shifts; the telemetry harness already infers this from the dataset’s class count.

J Activation function catalogue

Activation	Definition	Phase-equivariant?
CReLU	$\sigma(z) = \text{ReLU}(\text{Re } z) + i \text{ReLU}(\text{Im } z)$	no
ZReLU	$\sigma(z) = z$ if $\angle z \in [0, \pi/2]$, else 0	no
ModReLU	$\sigma(z) = \text{ReLU}(z + b) \cdot e^{i\angle z}$	yes
ComplexCardioid	$\sigma(z) = \frac{1}{2}(1 + \cos \angle z) \cdot z$	no
Siglog	$\sigma(z) = z/(1 + z)$	yes
ComplexTanh	$\sigma(z) = \tanh(\text{Re } z) + i \tanh(\text{Im } z)$	no

Table 12: The six activations characterized in this paper, and whether they preserve global phase equivariance layer-by-layer (Section A). Equivariance does not predict the empirical CVNN-vs-real gap pattern; five of these activations (all except ComplexTanh) are used in the RadioML ablation, and Siglog’s instability despite equivariance is part of the mechanism evidence in Section 3.6.5.